

# Pierre Kawak, Ph.D.

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## Education & Training

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|--------------------------------------------------------------------------|----------------------------|-------------|
| University of Wisconsin, Madison (UW-Madison)                            | Postdoctoral Scholarship   | 2026 –      |
| <i>Advisor: Ying Li</i>                                                  |                            |             |
| University of South Florida (USF)                                        | Postdoctoral Scholarship   | 2022 – 2025 |
| <i>Advisor: David S. Simmons</i>                                         |                            |             |
| Brigham Young University (BYU)                                           | Ph.D. Chemical Engineering | 2017 – 2022 |
| <i>Advisor: Douglas R. Tree</i>                                          |                            |             |
| Dissertation: Simulation of Crystal Nucleation in a Polymer Melt         |                            |             |
| American University of Sharjah (AUS)                                     | M.S. Chemical Engineering  | 2015 – 2017 |
| <i>Advisor: Ghaleb A. Hussein</i>                                        |                            |             |
| Dissertation: Ultrasound Triggered Release of Estrone-Targeted Liposomes |                            |             |
| American University of Sharjah (AUS)                                     | B.S. Chemical Engineering  | 2010 – 2015 |
| Minor Economics                                                          |                            |             |

## Publications

### Peer-Reviewed Journal Articles

- [6] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Glassy interphases reinforce elastomeric nanocomposites by enhancing volume expansion under strain”. *Proceedings of the National Academy of Sciences* 123 (2026). DOI: [10.1073/pnas.2528108123](https://doi.org/10.1073/pnas.2528108123).
- [5] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites”. *ACS Macro Letters* 14 (2025). DOI: [10.1021/acsmacrolett.5c00442](https://doi.org/10.1021/acsmacrolett.5c00442).
- [4] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Central Role of Filler-Polymer Interplay in Nonlinear Reinforcement of Elastomeric Nanocomposites”. *Macromolecules* 57 (2024). DOI: [10.1021/acs.macromol.4c00489](https://doi.org/10.1021/acs.macromol.4c00489).
- [3] **Pierre Kawak**, Christopher Akiki, and Douglas R. Tree. “Effect of local chain stiffness on oligomer crystallization from a melt”. *Physical Review Materials* 8 (2024), p. 075606. DOI: [10.1103/PhysRevMaterials.8.075606](https://doi.org/10.1103/PhysRevMaterials.8.075606).
- [2] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. “Semiflexible oligomers crystallize via a cooperative phase transition”. *Journal of Chemical Physics* 155 (2021), p. 214902. DOI: [10.1063/5.0067788](https://doi.org/10.1063/5.0067788).
- [1] Najla M. Salkho, Vinod Paul, **Pierre Kawak**, Rute F. Vitor, Ana M. Martins, Mohammad Al Sayah, and Ghaleb A. Hussein. “Ultrasonically controlled estrone-modified liposomes for estrogen-positive breast cancer therapy”. *Artificial Cells, Nanomedicine, and Biotechnology* 46 (2018), pp. 462–472. DOI: [10.1080/21691401.2018.1459634](https://doi.org/10.1080/21691401.2018.1459634).

### Preprints

- [4] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. “Fracture of Elastomeric Nanocomposites Due to Filler Pillar Formation” (in preparation).

- [3] Douglas R. Tree and **Pierre Kawak**. “The Search for a Molecular-Level Understanding of Nucleation in Polymer Crystallization” (in preparation).
- [1] **Pierre Kawak**, William F. Drayer, Mark E. Mackura, and David S. Simmons. “Amorphous Molecular Dynamics Analysis Toolkit (AMDAT): A Software Package for Analysis of Simulations of Structure and Dynamics in Supercooled Liquids and Complex Fluids” (2026). DOI: [10.48550/arXiv.2602.05865](https://doi.org/10.48550/arXiv.2602.05865).

### Software

- [2] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Filled Elastomer Simulation Package (FESP)*. Version 1.0.0. 2026. DOI: [10.5281/zenodo.19362854](https://doi.org/10.5281/zenodo.19362854).
- [1] David Simmons, **Pierre Kawak**, William Drayer, and Mark Mackura. *Amorphous Molecular Dynamics Analysis Toolkit (AMDAT)*. Version 1.0.1. 2025. DOI: [10.5281/zenodo.17417166](https://doi.org/10.5281/zenodo.17417166).

### Datasets

- [6] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Glassy interphases reinforce elastomeric nanocomposites by enhancing volume expansion under strain*. Dataset. 2026. DOI: [10.17632/dg9jxp8sj6](https://doi.org/10.17632/dg9jxp8sj6).
- [5] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Young's modulus and Poisson's ratio versus temperature data from uniaxial extension molecular dynamics simulations of model elastomers*. Dataset. 2026. DOI: [10.17632/c4bnns89g2](https://doi.org/10.17632/c4bnns89g2).
- [4] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Bulk modulus versus temperature data from equilibrium molecular dynamics simulations of a model neat elastomer*. Dataset. 2026. DOI: [10.17632/rfrb249t77](https://doi.org/10.17632/rfrb249t77).
- [3] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 1 - Render of a filled elastomer configuration*. Dataset. 2026. DOI: [10.17632/vrd38jc7hp](https://doi.org/10.17632/vrd38jc7hp).
- [2] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 2 - Relaxation time versus temperature*. Dataset. 2026. DOI: [10.17632/djm3kdvxwp](https://doi.org/10.17632/djm3kdvxwp).
- [1] **Pierre Kawak**, Harshad Bhapkar, and David Simmons. *Dataset for Origin of Heating-Induced Softening and Enthalpic Reinforcement in Elastomeric Nanocomposites - Figure 3 - Young's modulus, bulk modulus, and Poisson's ratio versus temperature with validation of Poisson's-ratio-mismatch theory*. Dataset. 2026. DOI: [10.17632/4bs9sxxb5](https://doi.org/10.17632/4bs9sxxb5).

## Awards & Fellowships

### Research Grants & Fellowships

|                                                                                                    |             |
|----------------------------------------------------------------------------------------------------|-------------|
| National Postdoctoral Association (NPA) <a href="#">IMPACT Fellowship</a>                          | 2023 – 2024 |
| One of six selected out of 100 applicants nationwide for funding & mentorship of proposed project. |             |
| National Science Foundation (NSF) CoPI Discover <a href="#">ACCESS</a> Compute Resource Grant      | Nov. 2023   |
| Awarded NSF funding for access to high performance computing resources.                            |             |
| American Physical Society (APS) <a href="#">Career Mentoring Fellowship</a>                        | 2023        |
| Received mentorship training, administered career talk at USF, & judged young trainee talks.       |             |

### Program Acceptance & Participation

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|----------------------------------------------------------------------------------------|-------------|
| <a href="#">Torrey Pines Foundations of Leadership Development Program</a> Participant | 2024 – 2025 |
| <a href="#">NSF &amp; SACNAS Grant Writing &amp; Peer Review Workshop</a> Attendance   | Aug. 2023   |

|                                                                                         |           |
|-----------------------------------------------------------------------------------------|-----------|
| Future Faculty Workshop Diverse Leaders for the Future Workshop Attendance              | June 2023 |
| Out in Science, Tech., Engineering, & Maths Professional Development Summit Participant | Mar. 2021 |
| UCSD SDSC High Performance Computing Summer Institute Attendee                          | Jul. 2018 |

### Conference Awards

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|---------------------------------------------------------------------------|-----------|
| Outstanding Poster Award at Gordon Research Conference on Polymer Physics | July 2024 |
| USF Annual Postdoctoral Research Symposium Best Poster Award \$200        | Mar. 2023 |
| APS Forum on International Physics Distinguished Student Award            | Fall 2022 |

### Excellence, Leadership & Service Awards

|                                                                       |             |
|-----------------------------------------------------------------------|-------------|
| AUS College of Engineering Hall of Fame Inductee                      | 2023        |
| BYU Chemical Engineering Department Graduate Student of the Month     | Sept. 2022  |
| BYU University Accessibility Center Banquet Scholarship Award \$1,500 | Fall 2021   |
| BYU Graduate Student Society Professional Presentation Award \$500    | Fall 2021   |
| BYU Chemical Engineering Department Travel Award                      | Fall 2021   |
| Delta Alpha Pi (DAPI) International Honor Society Inductee            | 2021        |
| AUS Biomedical Engineering Symposium Best Overall Talk Award \$700    | Fall 2016   |
| AUS 3× Dean's List for Academic Excellence                            | 2013 – 2014 |

## Research Mentorship Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------|
| <b>Amanda Sharrer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Ph.D. Candidate, USF</i>          | 2024-2025 |
| <ul style="list-style-type: none"> <li>– Onboarded and provided ongoing technical guidance for a project on molecular sequence selection for additive compatibilizers in mixed waste plastics.</li> <li>– Assisted in building molecular configurations and simulation protocols (equilibration/production, validation).</li> <li>– Supported analysis and visualization workflows using Python and OVITO to probe interfacial phenomena.</li> </ul>       |                                      |           |
| <b>Luis Zepeda</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Ph.D. Candidate, USF</i>          | 2024-2025 |
| <ul style="list-style-type: none"> <li>– Provided technical guidance for simulations of thin polymer films near supported and free surfaces.</li> <li>– Helped establish reliable simulation protocols to study structure and dynamics near interfaces.</li> <li>– Supported Python post-processing and visualization to extract interfacial trends and communicate results clearly.</li> </ul>                                                            |                                      |           |
| <b>Harshad Bhapkar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Ph.D. Candidate, USF</i>          | 2022-2025 |
| <ul style="list-style-type: none"> <li>– Directly supervised day-to-day research activities focused on mechanical response of polymer nanocomposites.</li> <li>– Provided hands-on mentorship spanning simulation planning, troubleshooting, analysis strategy, and interpretation of mechanical/structural metrics.</li> <li>– Served as a primary technical point-of-contact to maintain momentum, prioritize tasks, and ensure method rigor.</li> </ul> |                                      |           |
| <b>Makayla Branham</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Ph.D. Candidate, USF</i>          | 2023-2025 |
| <ul style="list-style-type: none"> <li>– Directly mentored and collaborated on research studying copolymer sequence effects on glass transition behavior.</li> <li>– Guided analysis, planning and interpretation to connect sequence-level descriptors with relaxation metrics.</li> </ul>                                                                                                                                                                |                                      |           |
| <b>Bao Ma</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Ph.D. Candidate, USF</i>          | 2023-2025 |
| <ul style="list-style-type: none"> <li>– Onboarded and provided technical guidance for thin-film polymer simulations near interfaces.</li> <li>– Supported development of simulation protocols and Python analysis for interfacial structure and dynamics.</li> </ul>                                                                                                                                                                                      |                                      |           |
| <b>Alyna Williams</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Undergraduate Researcher, USF</i> | 2024-2025 |
| <ul style="list-style-type: none"> <li>– Mentored end-to-end research execution: scientific computing, HPC usage, simulation workflows, and polymer physics literature grounding.</li> <li>– Co-developed generalized relaxation characterization pipeline across diverse molecular systems.</li> </ul>                                                                                                                                                    |                                      |           |

- Provided career mentorship and wrote letters of reference in support of professional opportunities.

**Tianna Virgo** *Undergraduate Researcher, USF* 2024

- Onboarded into simulation-based polymer physics research with structured guidance in scientific computing workflows, high-performance computing procedures, and best practices.

**Dr. Peijing Yue** *Ph.D. Candidate, USF* 2022-2024

- Provided sustained professional development/guidance during the Ph.D. to industry transition.
- Helped translate research strengths into industry-ready narratives and practical application materials.

**Annelise Curtin** *M.S. Student, USF* 2024

- Provided career guidance for graduate-school prep; placed as Ph.D. at the University of Minnesota.

**Austin Hartley** *Undergraduate Researcher, USF* 2023

- Provided career guidance for graduate-school prep; placed as Ph.D. at the Penn State University.

**Dr. William F. Drayer** *Ph.D. Candidate, USF* 2022-2023

- Provided professional development and career guidance during transition from Ph.D. to postdoc.

**Christopher Akiki** *Undergraduate Researcher, BYU* 2020-2022

- Co-authored a peer-reviewed manuscript on sensitivity of polymer crystallization to force-field choices.
- Collaborated on development of the Order-Parameter Wang–Landau (OPWL) simulation engine in C++, a generalized rare-event sampler for free energies and densities of states.
- Provided technical guidance on literature review and methodological approaches to polyethylene crystallization.
- Provided career guidance for graduate-school prep; placed as Ph.D. at University of Minnesota.

**Dr. Dakota S. Banks** *Undergraduate Researcher, BYU* 2018-2019

- Co-authored a peer-reviewed manuscript on minimum free energy path for polymer crystal nucleation.
- Collaborated on the design and implementation of a GPU-accelerated Monte Carlo Polymer Crystallization (MCPC) simulation engine in C++/CUDA.
- Provided career guidance for graduate-school prep; placed as Ph.D. at Brigham Young University.

**Beverly S. Delgado** *Undergraduate Researcher, BYU* 2019

- Supported onboarding into molecular simulations and polymer physics with literature and resources.

**Andrew S. Gibson** *Undergraduate Researcher, BYU* 2018

- Collaborated on development of the Monte Carlo Polymer Crystallization (MCPC) simulation engine in C++/CUDA and provided onboarding training in simulation workflows.

**Paul Kawak** *M.S. Student, AUS* 2015-2017

- Provided onboarding/mentorship on a project developing albumin-targeted smart nanocarriers for cancer treatment (synthesis, validation, and release studies probing ultrasonic-triggered response).

## Teaching Experience

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|--------------------------------------------------------------|---------------------------------------|------------------|
| Graduate Teaching Assistant                                  | <b>Thermodynamics</b>                 | Winter 2021      |
| <i>Brigham Young University</i>                              | <b>Separations Engineering</b>        | Fall 2021        |
|                                                              | <b>Heat &amp; Mass Transfer</b>       | 2018 – 2021 (3x) |
|                                                              | <b>Process Dynamics &amp; Control</b> | Fall 2018        |
| Volunteer Course Instructor; <i>University of the People</i> | <b>College Algebra</b>                | Spring 2018      |
| Graduate Instructor; <i>American University of Sharjah</i>   | <b>Principles of ChemE</b>            | 2016 – 2017 (3x) |
| Graduate Teaching Assistant                                  | <b>Corrosion Lab</b>                  | 2016 – 2017 (2x) |

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|---------------------------------------------------------------------------|-------------------------------------------|------------------|
| <i>American University of Sharjah</i>                                     | <b>ChemE Lab I</b>                        | 2015 – 2016 (2x) |
|                                                                           | <b>Graduate Desalination</b>              | Spring 2015      |
|                                                                           | <b>Wastewater Treatment</b>               | Spring 2015      |
| Undergraduate Teaching Assistant<br><i>American University of Sharjah</i> | <b>Mass Transfer</b>                      | 2014 – 2015 (3x) |
|                                                                           | <b>Kinetics</b>                           | Fall 2014        |
|                                                                           | <b>Thermodynamics</b>                     | Spring 2014      |
| Private Tutor                                                             | <b>Maths, Engineering, Business, etc.</b> | 2008 – now       |

## Community & Service

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### Peer Review

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|-------------------------------------------------------------------------------------------|------------|
| American Chemical Society <a href="#">Macromolecules</a>                                  | 9× Reviews |
| American Chemical Society <a href="#">Journal of Chemical Information and Modeling</a>    | 2× Reviews |
| American Chemical Society <a href="#">Petroleum Research Fund (PRF)</a>                   | 2× Review  |
| Freiburg Institute for Advanced Studies <a href="#">Early Career Fellowship Programme</a> | 1× Review  |
| Wiley <a href="#">Journal of Polymer Science</a>                                          | 1× Review  |

### Professional Society & Conference Leadership

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|-----------------------------------------------------------------------------------------------------------|------------------------|
| Co-Chair, Gordon Research Seminar on Polymer Physics                                                      | 2026                   |
| Organized technical sessions and networking; fundraised for the seminar.                                  |                        |
| <i>American Physical Society (APS)</i>                                                                    |                        |
| Session Chair & Organizer: “Polymer Structure & Dynamics across Length & Time Scales”                     | 2024, 2025, 2026, 2027 |
| Session Chair & Organizer: “Tuning Polymer Properties Through Macromolecular Architecture & Interactions” | 2025, 2026, 2027       |
| Member of <a href="#">Committee on International Freedom of Scientists</a>                                | 2025-2026              |
| Member of Division of Polymer Physics (DPOLY) Membership Committee                                        | 2024-2025              |
| Organizer & Winner of Inaugural DPOLY T-Shirt Design Competition                                          | Mar 2024               |
| Ranked Undergraduate Talks & Posters                                                                      | 2022 – 2023            |
| Session Chair “ <a href="#">Polymers &amp; Composites for Energy Storage &amp; Conversion I</a> ”         | Mar 2023               |
| <i>Early Career Researchers in Polymer Physics</i>                                                        |                        |
| Administrator of 550 member Slack channel for collaboration & networking                                  | 2022 – present         |
| Organizer, <a href="#">Virtual Polymer Physics Symposium</a> (150+ attendees)                             | Aug. 2023              |
| Cofounder & Organizer of <a href="#">Self-Development Seminar</a> series                                  | 2022 – present         |
| <i>Out in Science Technology Engineering &amp; Mathematics (oSTEM), Inc.</i>                              |                        |
| Represented oSTEM at MAA MathFest outreach tables.                                                        | Aug. 2023              |
| Annual Conference Volunteer & Organizer                                                                   | Nov. 2022              |
| Annual Conference Merchandise Team Organizer                                                              | Nov. 2022              |

### Departmental & Institutional Service

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|---------------------------------------------------------------------------------|----------------|
| <i>UW–Madison Postdoctoral Association</i>                                      |                |
| Executive Committee Treasurer                                                   | 2026 – present |
| Fundraised for the 2026 Postdoctoral Research Symposium.                        | 2026           |
| <i>University of South Florida (USF) Postdoctoral Scholar Association (PSA)</i> |                |
| Founded & Chaired PSA Executive Committee at USF serving 200 postdocs           | 2023 – present |

|                                                                                                                                                   |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Organized Inaugural ELEVATE Talk Series                                                                                                           | 2024 – 2025      |
| Led Postdoc Highlight Interviews, Socials, & Orientations                                                                                         | 2023 – present   |
| <i>BYU Chemical Engineering Graduate Student Council (GSC)</i>                                                                                    |                  |
| President & Cofounder                                                                                                                             | 2018 – 2021      |
| Prepared onboarding materials and resources for new graduate students.                                                                            |                  |
| Created graduate-student health and well-being surveys, including a financial-health survey administered in fall 2021.                            |                  |
| Organizer of Department Recruitment Poster Event                                                                                                  | 2019, 2020, 2021 |
| Department BBQ Social Organizer                                                                                                                   | 2018 – 2021      |
| Social Media Accounts Manager                                                                                                                     | Fall 2021        |
| <b>Outreach, Inclusion &amp; Public Engagement</b>                                                                                                |                  |
| Advisory Board Member, <a href="#">Pulse Physics</a>                                                                                              | 2026 – present   |
| High-school physics club network.                                                                                                                 |                  |
| Presidential AI Challenge — Volunteer Judge & Mentor                                                                                              | 2025             |
| Posted machine-learning project guidance for parents and students across the U.S.; hosted office hours and judged inaugural-session entries.      |                  |
| Volunteer Judge, State Science & Engineering Fair of Florida (SSEF)                                                                               | 2023, 2024       |
| Judged several Utah science fairs during Ph.D. training at BYU.                                                                                   |                  |
| <i>American Chemical Society (ACS)</i>                                                                                                            |                  |
| <a href="#">Science Coaches</a>                                                                                                                   | 2023 – 2024      |
| Supported a Bradenton Christian School teacher with questions about her physics coursework by email and phone.                                    |                  |
| Bradenton Christian School, Florida — Full-Day Visit                                                                                              | Mar. 2024        |
| Discussed materials-science careers with around 100 grade 8–10 students.                                                                          |                  |
| Presented “Academia, Polymers, and the Ocean,” engaging students in a sustainability discussion that prompted ideas for addressing plastic waste. |                  |
| <i>American Physical Society (APS)</i>                                                                                                            |                  |
| Member of <a href="#">National Mentoring Community</a>                                                                                            | 2022 – present   |
| <a href="#">Physicists To-Go</a> Public Engagement Program Participant                                                                            |                  |
| Josephine C. Locke Elementary School, Chicago — Scholar Visit                                                                                     | Oct. 2023        |
| Spent a full day teaching and engaging with about 150 grade-8 students.                                                                           |                  |
| <i>University of South Florida (USF)</i>                                                                                                          |                  |
| Martinez Middle School — Great American Teach-In                                                                                                  | Nov. 2024        |
| Delivered a talk on careers in physics.                                                                                                           |                  |
| Facilitated a summer workshop for high-school students on programming and scientific computing.                                                   | June 2023        |
| <i>American Society for Engineering Education (ASEE)</i>                                                                                          |                  |
| Facilitator of <a href="#">Trans Allyship Safe Zone Ally Training Workshop</a>                                                                    | Mar. 2023        |
| Member of ASEE LGBTQ+ Advocacy in STEM <a href="#">Virtual Community of Practice</a>                                                              | 2022 – present   |
| <i>Out in Science Technology Engineering &amp; Mathematics (oSTEM), Inc.</i>                                                                      |                  |
| Scholarship Coordinator; led applicant reviews.                                                                                                   | 2022 – 2023      |
| Mentorship Program Volunteer                                                                                                                      | 2021 – present   |

## Technical Presentations

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- [29] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Low-Strain Reinforcement as a Volume-Expansion Effect in Filled Elastomers with Glassy Interphases". Global Physics Summit. American Physical Society. Denver, CO, 2026.
- [28] **Pierre Kawak**. "Molecular Simulations and Machine Learning for Sustainable Polymer Innovation". Southeast Polymer Forum. University of Georgia, Athens. Athens, GA, 2025.
- [27] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Exploring the Role of Polymer-Filler Interactions in Modulating Elastomeric Reinforcement". Global Physics Summit. American Physical Society. Anaheim, CA, 2025.
- [26] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Tuning Polymer-Filler Interactions to Modulate Elastomeric Reinforcement". Rubber Division Spring Technical Meeting. American Chemical Society. Lake Buena Vista, FL, 2025.
- [25] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Tuning Reinforcement, Void Formation, and Fracture in Elastomeric Nanocomposites: Toward High-Performance, Sustainable Tire Materials". Chemical, Biological, & Materials Engineering Department Seminar. University of South Florida. Tampa, FL, 2025.
- [24] **Pierre Kawak**. "Molecular Simulations for Greener Polymers: From Theory to Reality". Annual Meeting. American Institute of Chemical Engineers. San Diego, CA, 2024.
- [23] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Contrasting Reinforcement Mechanisms in Elastomeric Nanocomposites". Annual Meeting. American Institute of Chemical Engineers. San Diego, CA, 2024.
- [22] **Pierre Kawak**. "Filler-Filler Contacts Reinforce Filled Elastomers at High Strains". GRC Polymer Physics. Gordon Research Conferences. South Hadley, MA, 2024.
- [21] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Filler-Filler Contacts Reinforce Filled Elastomers at High Strains". Annual Postdoctoral Research Symposium. University of South Florida. Tampa, FL, 2024.
- [20] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Polymer-Filler Competition-Driven Reinforcement Beyond the Payne Effect in Elastomeric Nanocomposites". March Meeting. American Physical Society. Minneapolis, MN, 2024.
- [19] Harshad Bhapkar, **Pierre Kawak**, and David S. Simmons. "Exploring the Effects of Nanoparticle Loading, Dispersion and Structure on the Stress Response of Elastomeric Nanocomposites". March Meeting. American Physical Society. Minneapolis, MN, 2024.
- [18] **Pierre Kawak**, David S. Simmons, and Douglas R. Tree. "Rational Sustainable Polymer Materials Design Using Multiscale Simulation and Theory". Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [17] **Pierre Kawak**, Makayla Branham, William F. Drayer, and David S. Simmons. "Tuning Polymer Dynamics Via Sequence Control". Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [16] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Elucidating the Molecular Origins of Reinforcement in Filled Elastomers Via Spatial- and Species-Resolved Stresses from Molecular Dynamics Simulations". Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.
- [15] Harshad Bhapkar, **Pierre Kawak**, and David S. Simmons. "Insights into the Dependence of Elastomeric Nanocomposite Mechanics on Nanoparticulate Properties". Annual Meeting. American Institute of Chemical Engineers. Orlando, FL, 2023.

- [14] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Dissecting the Payne Effect: How Filler-Polymer Competition Reinforces Elastomeric Nanocomposites". IOP Polymer Physics Group Graduate Symposium. Institute of Physics. Virtual, 2023.
- [13] **Pierre Kawak**. "Career Paths in Physics". Physics Colloquia Series. University of South Florida Department of Physics. Tampa, FL, 2023.
- [12] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Exploring Mechanisms of Enhanced Dissipation in Nanoparticle-filled Rubber Using Molecular Dynamics". Annual Postdoctoral Research Symposium. University of South Florida. Tampa, FL, 2023.
- [11] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Exploring mechanisms of enhanced dissipation in nanoparticle-filled rubber using molecular dynamics". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [10] Douglas R. Tree and **Pierre Kawak**. "Free Energy Analysis of Crystal Nucleation of Semiflexible Polymers". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [9] **Pierre Kawak**, Harshad Bhapkar, and David S. Simmons. "Spatially resolving energy dissipation in molecular dynamics of polymer nanocomposites". March Meeting. American Physical Society. Las Vegas, NV, 2023.
- [8] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Acute Sensitivity of Polymer Crystallization Phase Behavior to Intermolecular Interactions". Annual Meeting. American Institute of Chemical Engineers. Phoenix, AZ, 2022.
- [7] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Free Energy Analysis of Polymer Crystal Nucleation Indicates Cooperative Crystallization and Nematic Alignment". March Meeting. American Physical Society. Chicago, IL, 2022.
- [6] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "Free Energy Surfaces for Homogeneous Nucleation in a Polymer Melt". Annual Meeting. American Institute of Chemical Engineers. Boston, MA, 2021.
- [5] **Pierre Kawak**, Dakota S. Banks, and Douglas R. Tree. "GPU-accelerated Wang-Landau Simulation of Polymer Crystallization". March Meeting. American Physical Society. Virtual, 2021.
- [4] **Pierre Kawak**, Andrew S. Gibson, Logan S. Brown, Beverly Delgado, Douglas R. Tree, and Dakota S. Banks. "Investigating Primary Nucleation in Polymer Melts using GPU-Accelerated Wang-Landau Simulations". Annual Meeting. American Institute of Chemical Engineers. Virtual, 2020.
- [3] **Pierre Kawak**, Andrew S. Gibson, Logan S. Brown, Beverly Delgado, and Douglas R. Tree. "Wang-Landau Simulation of the Free Energy Surface of Crystallization in a Polymer Melt". March Meeting. American Physical Society. Virtual, 2020.
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